Practical – 6.1

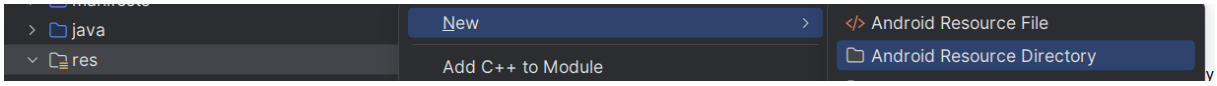
AIM

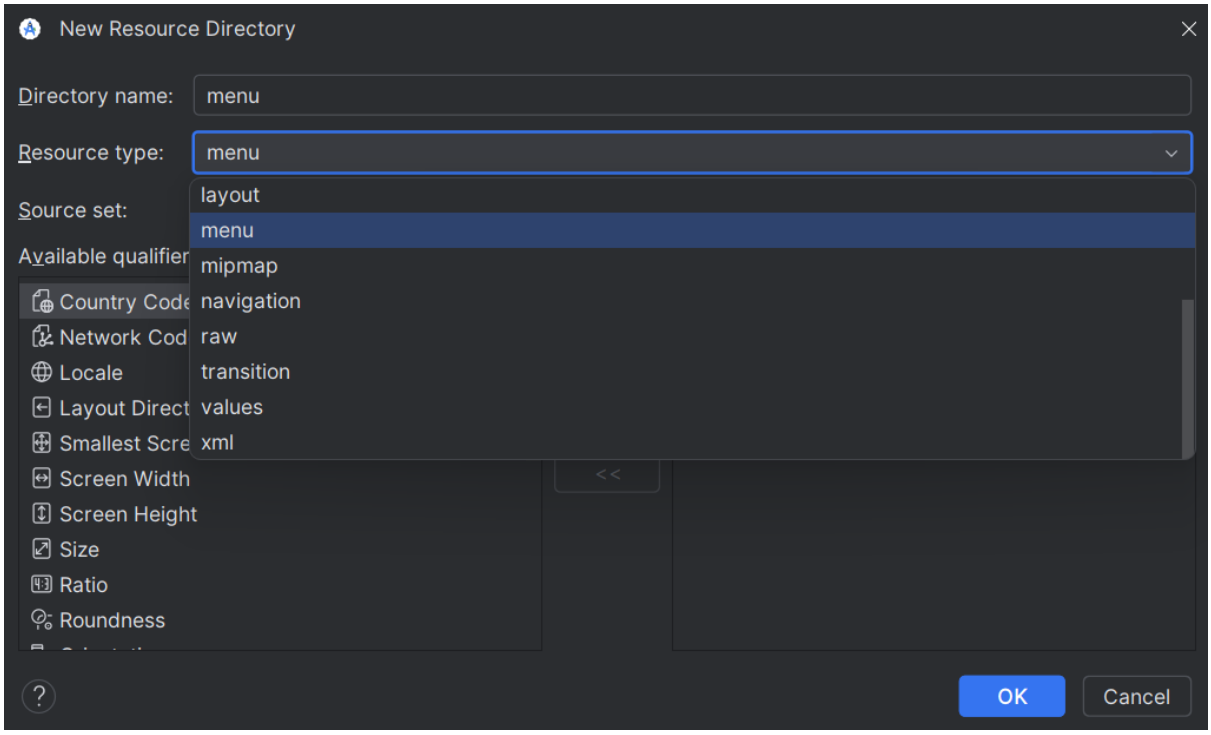
To implement an Android application demonstrating **programming of menus and dialogs** for user interaction.

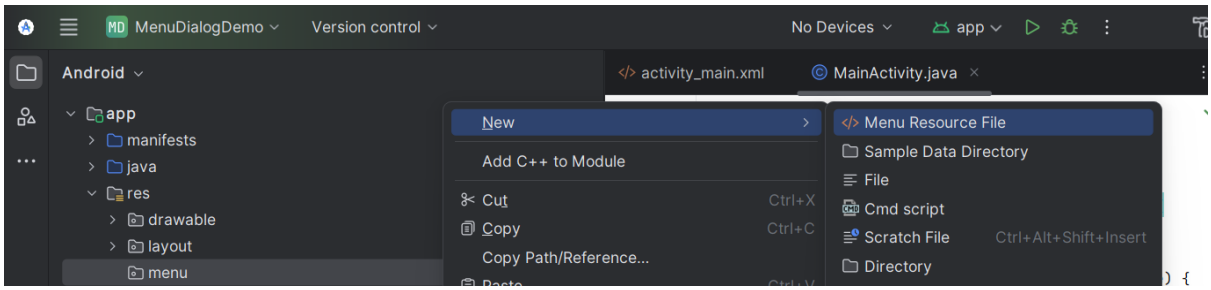
PROBLEM STATEMENT

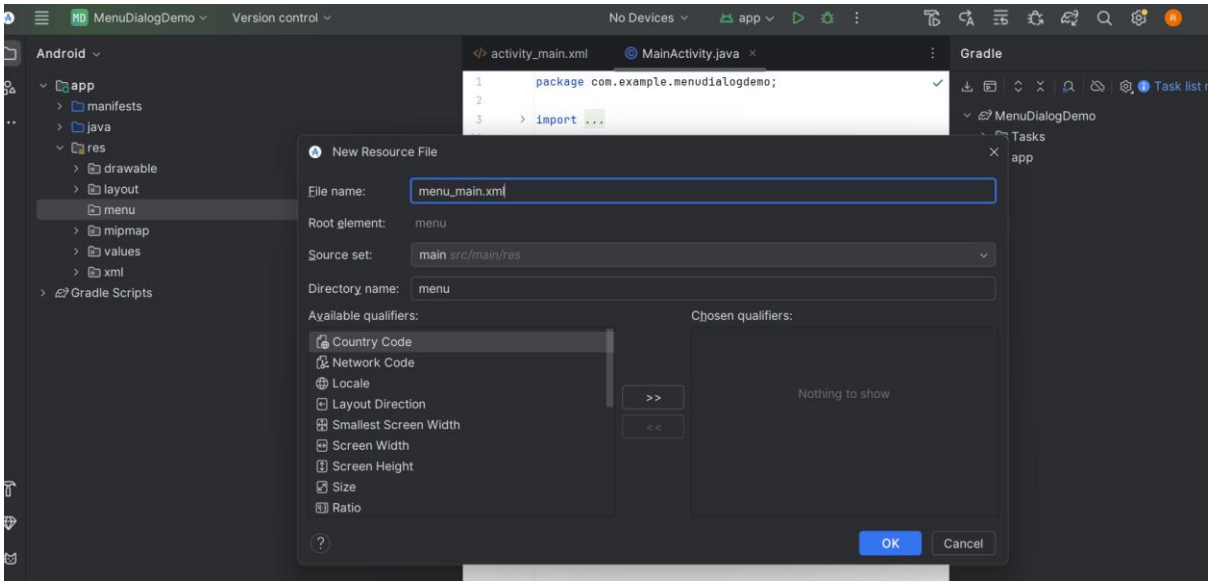
Design and develop an Android application that uses an **Options Menu** with multiple menu items. On selecting a menu item, display an appropriate **dialog** to show information or to confirm user action, and perform the required operation based on the user's response.

```
app
 ├── java
 |   └── com.example.menudialogapp
 |       └── MainActivity.java
 ├── res
 |   ├── layout
 |   |   └── activity_main.xml
 |   ├── menu
 |   |   └── menu_main.xml
 |   └── values
 |       └── strings.xml
```




New Resource Directory
Directory name:
menu
Resource type:
menu
Source set:
Available qualifier
layout
menu
mipmap
navigation
raw
transition
values
xml
Country Code
Network Cod
Locale
Layout Direct
Smallest Scre
Screen Width
Screen Height
Size
Ratio
Roundness
OK
Cancel


MenuDialogDemo
Version control
No Devices
app
Android
activity_main.xml
MainActivity.java
app
manifests
java
res
drawable
layout
menu
New
Add C++ to Module
Cut
Ctrl+X
Copy
Ctrl+C
Copy Path/Reference...
Menu Resource File
Sample Data Directory
File
Cmd script
Scratch File
Ctrl+Alt+Shift+Insert
Directory


MenuDialogDemo
Version control
No Devices
app
Android
activity_main.xml
MainActivity.java
Gradle
package com.example.menudialogdemo;
import ...
app
manifests
java
res
drawable
layout
menu
mipmap
values
xml
Gradle Scripts
MenuDialogDemo
Tasks
app
New Resource File
File name:
menu_main.xml
Root element:
menu
Source set:
main src/main/res
Directory name:
menu
Available qualifiers:
Country Code
Network Code
Locale
Layout Direction
Smallest Screen Width
Screen Width
Screen Height
Size
Ratio
Chosen qualifiers:
Nothing to show
OK
Cancel

activity_main.xml:SourceCode:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
    <!-- Toolbar -->
    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="?attr/colorPrimary"
        android:title="Menu Dialog App"
        android:titleTextColor="@android:color/white"
        app:popupTheme="@style/ThemeOverlay.AppCompat.Light"/>
    <!-- Content -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:orientation="vertical">
        <TextView
            android:id="@+id/txtMessage"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Select an option from menu"
            android:textSize="20sp"
            android:textStyle="bold"/>
    </LinearLayout>
```

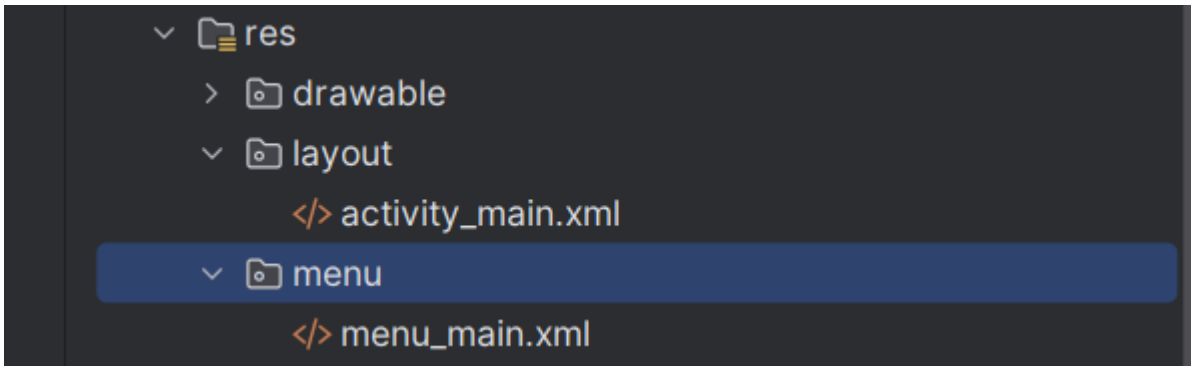
```
</LinearLayout>
```

MainActivity.java

```
package com.example.menudialogapp;
import android.content.DialogInterface;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.TextView;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
public class MainActivity extends AppCompatActivity {
    TextView txtMessage;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Connect Toolbar to Action Bar
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
        // Connect TextView
        txtMessage = findViewById(R.id.txtMessage);
    }
    // Inflate Options Menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }
    // Handle Menu Item Clicks
    @Override
```

```
    public boolean onOptionsItemSelected(MenuItem item) {
        int itemId = item.getItemId();
        if (itemId == R.id.menu_about) {
            showAboutDialog();
            return true;
        }
        if (itemId == R.id.menu_reset) {
            showResetDialog();
            return true;
        }
        if (itemId == R.id.menu_exit) {
            showExitDialog();
            return true;
        }
        return super.onOptionsItemSelected(item);
    }
    // About Dialog
    private void showAboutDialog() {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setTitle("About App");
        builder.setMessage("This application demonstrates the use of Options Menu and Alert Dialogs in
Android.");
        builder.setPositiveButton("OK", null);
        builder.show();
    }
    // Reset Confirmation Dialog
    private void showResetDialog() {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setTitle("Confirm Reset");
        builder.setMessage("Do you want to reset the displayed text?");
        builder.setPositiveButton("Yes", new DialogInterface.OnClickListener() {
```

```
            @Override
            public void onClick(DialogInterface dialog, int which) {
                txtMessage.setText("Text has been reset");
            }
        });
        builder.setNegativeButton("No", null);
        builder.show();
    }
    // Exit Confirmation Dialog
    private void showExitDialog() {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setTitle("Exit Application");
        builder.setMessage("Are you sure you want to exit?");
        builder.setPositiveButton("Yes", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int which) {
                finish();
            }
        });
        builder.setNegativeButton("No", null);
        builder.show();
    }
}
```



Menu_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item
      android:id="@+id/menu_about"
      android:title="About App"/>
   <item
      android:id="@+id/menu_reset"
      android:title="Reset Text"/>
   <item
      android:id="@+id/menu_exit"
      android:title="Exit App"/>
</menu>
```